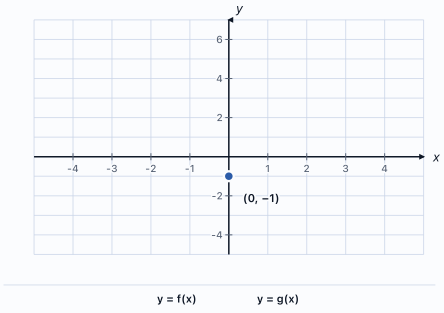
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Coordinate Geometry & Linear Programming Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 12x + 50$ in the domain of Coordinate Geometry & Linear Programming. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(8)$.



A 38
B 18
C 30
D 32

2 MCQ

A mathematical model in Coordinate Geometry & Linear Programming is defined by the cubic function $y_2(x) = 2x^3 - 6x^2 + 15x - 11$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 33
B 69
C 18
D 21

3 MCQ

In Coordinate Geometry & Linear Programming, two displacement vectors are given by $\mathbf{u}_3 = \langle 6, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -2
B 98
C 100
D 210

4 MCQ

The linear growth profile shown in the diagram for Coordinate Geometry & Linear Programming follows an arithmetic progression with first term $a_1 = 7$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 91

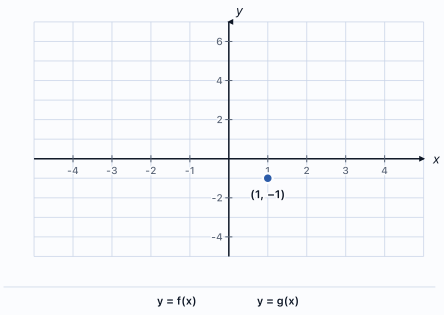
B 825

C 735

D 1365

5 MCQ

In Coordinate Geometry & Linear Programming, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 24x)\,\mathrm{d}x$.



A 120

B $120 + 1$

C 49

D 96

6 MCQ

In Coordinate Geometry & Linear Programming, consider the quadratic function $f_6(x) = x^2 - 14x + 135$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(14)$.

A 221

B 228

C 135

D 214

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 7x^2 + 33x - 18$ studied in Coordinate Geometry & Linear Programming. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 417

B 305

C 272

D 67

8 MCQ

In Coordinate Geometry & Linear Programming, two displacement vectors are given by $\mathbf{u}_8 = \langle 7, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 21

B 98

C 119

D 189

9 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 2)$ and $B(2, 8)$ (Segment #1).

A $y = -\frac{1}{3}x + 9$

B $y = \frac{1}{3}x - 2$

C $y = -\frac{1}{3}x + \frac{16}{3}$

D $y = 3x + 5$

10 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 4)$ and $B(6, 12)$ (Segment #2).

A $y = \frac{1}{2}x - 5$

B $y = -\frac{1}{2}x + 10$

C $y = -\frac{1}{2}x + 15$

D $y = 2x + 8$

11 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 6)$ and $B(10, 8)$ (Segment #3).

A $y = 3x - 8$

B $y = -3x + 17$

C $y = \frac{1}{3}x + 7$

D $y = -3x + 28$

12 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(6, 8)$ and $B(14, 12)$ (Segment #4).

A $y = -2x + 23$

B $y = -2x + 30$

C $y = \frac{1}{2}x + 10$

D $y = 2x - 11$

13 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(8, 10)$ and $B(10, 16)$ (Segment #5).

A $y = \frac{1}{3}x - 10$

B $y = -\frac{1}{3}x + 16$

C $y = 3x + 13$

D $y = -\frac{1}{3}x + 25$

14 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(10, 0)$ and $B(14, 8)$ (Segment #6).

A $y = -\frac{1}{2}x + 19$

B $y = -\frac{1}{2}x + 10$

C $y = 2x + 4$

D $y = \frac{1}{2}x - 13$

15 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(12, 2)$ and $B(18, 4)$ (Segment #7).

A $y = -3x + 21$

B $y = 3x - 16$

C $y = -3x + 48$

D $y = \frac{1}{3}x + 3$

16 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 4)$ and $B(8, 8)$ (Segment #8).

A $y = 2x - 5$

B $y = \frac{1}{2}x + 6$

C $y = -2x + 14$

D $y = -2x + 13$

17 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 6)$ and $B(4, 12)$ (Segment #9).

A $y = 3x + 9$

B $y = \frac{1}{3}x - 4$

C $y = -\frac{1}{3}x + 15$

D $y = -\frac{1}{3}x + 10$

18 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 8)$ and $B(8, 16)$ (Segment #10).

A $y = -\frac{1}{2}x + 15$

B $y = \frac{1}{2}x - 7$

C $y = -\frac{1}{2}x + 21$

D $y = 2x + 12$

19 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(6, 10)$ and $B(12, 12)$ (Segment #11).

A $y = 3x - 10$

B $y = -3x + 23$

C $y = -3x + 38$

D $y = \frac{1}{3}x + 11$

20 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(8, 0)$ and $B(16, 4)$ (Segment #12).

A $y = \frac{1}{2}x + 2$

B $y = -2x + 26$

C $y = 2x - 13$

D $y = -2x + 17$

21 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(10, 2)$ and $B(12, 8)$ (Segment #13).

A $y = -\frac{1}{3}x + 19$

B $y = 3x + 5$

C $y = \frac{1}{3}x - 12$

D $y = -\frac{1}{3}x + \frac{26}{3}$

22 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(12, 4)$ and $B(16, 12)$ (Segment #14).

A $y = 2x + 8$

B $y = -\frac{1}{2}x + 25$

C $y = -\frac{1}{2}x + 15$

D $y = \frac{1}{2}x - 15$

23 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(0, 6)$ and $B(6, 8)$ (Segment #15).

A $y = -3x + 13$

B $y = \frac{1}{3}x + 7$

C $y = 3x - 4$

D $y = -3x + 16$

24 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(2, 8)$ and $B(10, 12)$ (Segment #16).

A $y = -2x + 19$

B $y = \frac{1}{2}x + 10$

C $y = 2x - 7$

D $y = -2x + 22$

25 MCQ

Find the equation of the perpendicular bisector of the line segment joining $A(4, 10)$ and $B(6, 16)$ (Segment #17).

A $y = -\frac{1}{3}x + 21$

B $y = -\frac{1}{3}x + \frac{44}{3}$

C $y = 3x + 13$

D $y = \frac{1}{3}x - 6$